

Topic

Predicting Credit Card Customer Default Risk Using Machine Learning

Business Problem

Banks and financial institutions generate substantial revenue from credit card services. When firms approve and issue credit cards, customers are generally considered to be in good financial condition. However, economic conditions and personal financial situations can change over time. People may lose jobs, experience divorce, or face other life events that negatively affect their ability to repay debt. Because of this, credit card issuers must continuously monitor customers' ability to manage their financial obligations.

Organizations that can correctly identify risky customers may improve their lending practices by adjusting credit limits, monitoring distressed accounts more closely, and reducing exposure to potential losses. Predicting default risk before serious delinquency occurs may help financial institutions remain financially stable while improving risk management practices.

This project will explore several research questions related to credit card default prediction. The study will examine which variables are the strongest predictors of default, whether machine learning models can accurately predict default risk, and which models perform best. Additional questions include whether gender, age, education level, and credit card limits have any relationship with default rates.

Dataset

For this project, the Default of Credit Card Clients Dataset (UCI_Credit_Card.csv) will be used. The data was obtained from the UCI Machine Learning Repository through Kaggle. This dataset is considered credible because it originates from a well-known academic source frequently used in machine learning and predictive analytics research.

The dataset contains 25 variables and approximately 30,000 customer records. Each row represents one credit card customer. Variables include demographic information such as credit limit, sex, education, marriage status, and age. The dataset also contains customer payment history, bill statement amounts, repayment amounts, and repayment status over several months. In addition, the dataset includes a target variable indicating whether a customer defaulted on their payment the following month. Monetary values in the dataset are recorded in New Taiwan Dollars (NT\$). The data was collected from credit card clients in Taiwan and reflects customer credit behavior within Taiwanese financial institutions.

Methods

Several analytical methods are expected to be used throughout this project. The project will begin with exploratory data analysis and data cleaning to better understand the structure and quality of the dataset. Correlation analysis and visualizations will also be performed to identify possible relationships between variables.

Machine learning classification techniques such as Logistic Regression and Random Forest are expected to be implemented to predict customer default risk. Model performance will be evaluated using several common evaluation metrics including accuracy, precision, recall, confusion matrix results, and ROC-AUC scores. These methods will help determine which model performs best for predicting credit card default.

Ethical Considerations

The dataset comes from a credible academic source and does not contain personally identifiable information such as customer names, account numbers, or addresses. Because of this, privacy risks are limited. However, ethical concerns still exist when machine learning models are used in financial decision-making.

One potential concern is the possibility of bias against certain demographic groups. Financial institutions must ensure that predictive models do not unfairly disadvantage individuals based on age, gender, education, or marital status. Another concern is relying too heavily on automated systems when making lending decisions. Machine learning models are not always fully accurate and may occasionally produce incorrect predictions. For this reason, predictive models should be used as tools to support human decision-making rather than fully replace human judgment.

Challenges / Issues

One challenge in this project is that customer financial behavior may differ across countries and cultures. Since this dataset was collected in Taiwan, customer spending habits and borrowing behavior may not fully represent customers in the United States or other countries. Economic conditions, cultural influences, and financial priorities may affect how customers use credit cards and manage debt.

Another challenge may involve interpreting encoded variables and addressing class imbalance within the dataset, since the number of non-default customers is larger than the number of default customers. Additional effort may also be required to avoid overfitting and ensure that the models generalize well.

References

UCI Machine Learning Repository: Default of Credit Card Clients Dataset

[Kaggle Credit Card Default Dataset](#)

Federal Reserve Bank of St. Louis. (2026, February 24). Delinquency Rate on Credit Card Loans, All Commercial Banks (DRCCLACBS).

[Federal Reserve Economic Data \(FRED\)](#)

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